

Lesson 5 · Timed practice

NCEA Level 2 · AS 91262 · Mixed questions

Time: 30 minutes

Name: _____

Date: _____

Exam conditions: no notes; calculator allowed. Show all working and write the derivative first. [A] Achieved, [M] Merit, [E] Excellence.

1. [A]

(a) Find the gradient of $f(x) = x^4 - 3x^2 + 2x$ at the point where $x = -1$.

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(b) Find the equation of the tangent to $y = 2x^2 - 5x + 1$ at $x = 3$.

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(c) $f'(x) = 3x^2 - 8x + 2$ and $f(2) = 1$. Find $f(x)$.

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(d) For which values of x is $f(x) = x^3 - 6x^2 + 5$ decreasing?

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2. [M]

(a) A particle moves with $s = 4t^3 - 15t^2 + 12t$ metres after t seconds. Find the times when it is at rest.

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(b) A drone takes off from the ground with vertical velocity $v = 6t - t^2 \text{ m s}^{-1}$ for $0 \leq t \leq 6$. Find its maximum height, and justify that it is the maximum.

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Question 3 [E]

- (iii) At another match, the number of spectators is modelled by the equation $P = kt^2 - 2t^4 + 750$, where k is a constant.

The number of spectators is growing fastest when $t = 4$.

Find the value of k .

Explain your reasoning clearly, using correct mathematical statements.